What is Git?

Git is a popular version control system.

It was created by Linus Torvalds in 2005, and has been maintained by Junio Hamano since then.

Git **is an open-source, local version control system (VCS), while GitHub is a web-based platform that hosts Git repositories online and provides collaboration tools**. You can use Git without GitHub, but you cannot use GitHub without using Git

It is used for:

- Tracking code changes
- Tracking who made changes
- Coding collaboration

Key Git Concepts

- **Repository:** A folder where Git tracks your project and its history.
- **Clone:** Make a copy of a remote repository on your computer.
- **Stage:** Tell Git which changes you want to save next.
- **Commit:** Save a snapshot of your staged changes.
- **Branch:** Work on different versions or features at the same time.
- **Merge:** Combine changes from different branches.
- **Pull:** Get the latest changes from a remote repository.
- **Push:** Send your changes to a remote repository.

Working with Git

- Initialize Git on a folder, making it a **Repository**
- Git now creates a hidden folder to keep track of changes in that folder
- When a file is changed, added or deleted, it is considered **modified**
- You select the modified files you want to **Stage**
- The **Staged** files are **Committed**, which prompts Git to store a **permanent** snapshot of the files
- Git allows you to see the full history of every commit.
- You can revert back to any previous commit.
- Git does not store a separate copy of every file in every commit, but keeps track of changes made in each commit!

What is GitHub?

GitHub is a cloud-based service that takes the core functionality of Git and enhances it with a graphical user interface and additional features for collaboration and project management. It serves as a central location to store your Git repositories on the web.

- Git is not the same as GitHub.
- GitHub makes tools that use Git.
- GitHub is the largest host of source code in the world, and has been owned by Microsoft since 2018.
- In this tutorial, we will focus on using Git with GitHub.

How To Install Git on Ubuntu 20.04

Last Updated : 23 Jul, 2025

- Git, a popular version control system, is widely used for managing code in software development projects. It tracks changes in code, allowing collaboration and easy reversion to previous versions if needed. This article will outline two methods for installing Git on your Ubuntu system.

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- Method 1 : Using APT package manager
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Method 1 : Using APT package manager

Step 1 : Update the package list

To make sure we are getting the latest version of git, update the package list using **apt update** command.

```
sudo apt update -y
```

Step 2 : Install Git

Once repositories are up to date, we will install git using **apt install** command.

```
sudo apt install git -y
```

Step 3 : Verify installation

Once, installation is done, verify installation using **git --version** command.

```
git --version
```

Below is a table of differences between Git and GitHub:

Git	GitHub
A software tool.	A web-based service.
A distributed version control system (DVCS).	A repository hosting and collaboration platform.
Primarily used through a command-line interface.	Provides a web-based graphical interface.
Installed locally on a developer’s system.	Hosted on the web.
Maintained by the open-source community.	Maintained by Microsoft.
Focuses on version control and change tracking.	Focuses on code hosting and collaboration.
Manages source code history.	Hosts version-controlled repositories.
First released in 2005.	Launched in 2008.
No built-in user access management	Includes user and access management.
Open-source licensed.	Offers free and paid plans.
Limited external tool integration.	Provides a marketplace for integrations.
Git competes with CVS, SVN, Mercurial, etc.	GitHub competes with GitLab, Bit Bucket, Azure DevOps Server, etc.